European Developer Conference



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Developing Bluetooth and Wifi apps

The Plan (1/2)

Bluetooth in a nutshell

Wifi in a nutshell

Pros & Cons

Bluetooth:
- **Architecture**
- **How to improve the discovery UI**
- **Telephony profiles**
- **How to implement the BIP profile**
- **How to implement the PAN profile**
- **How to implement the HID profile**

The Plan (2/2)

WiFi: - Palm WiFi SDK solution

- How to develop a WiFi application

- How to develop a WiFi HW solution

Bluetooth in the Future

WiFi in the Future

Bluetooth & WiFi : Best developers resources

Bluetooth in a nutshell (1/4)

Wireless link

Designed as a cable replacement

Targeted Devices:
- **Mobile phones**
- **Mobile PC's (laptops)**
- **Handheld devices**
- **Wireless peripherals**
  - Headsets
  - Cameras
- **Access Points:**
  - bridges to wired networks

Bluetooth in a nutshell (2/4)

Open standard

- Specification freely available
- www.bluetooth.com

Very focused on mobile wireless links

- Very small
- Low cost
- Low power consumption

Huge industry support

- Over 2,000 member companies

Bluetooth in a nutshell (3/4)

2.4 GHz Open band

- Specification covers both hardware and software
- Uses the open 2.4 GHz frequency range
- This frequency is available worldwide
- Frequency hopping and Time Division Multiplexing

10 - 100 meter range (“personal bubble”)

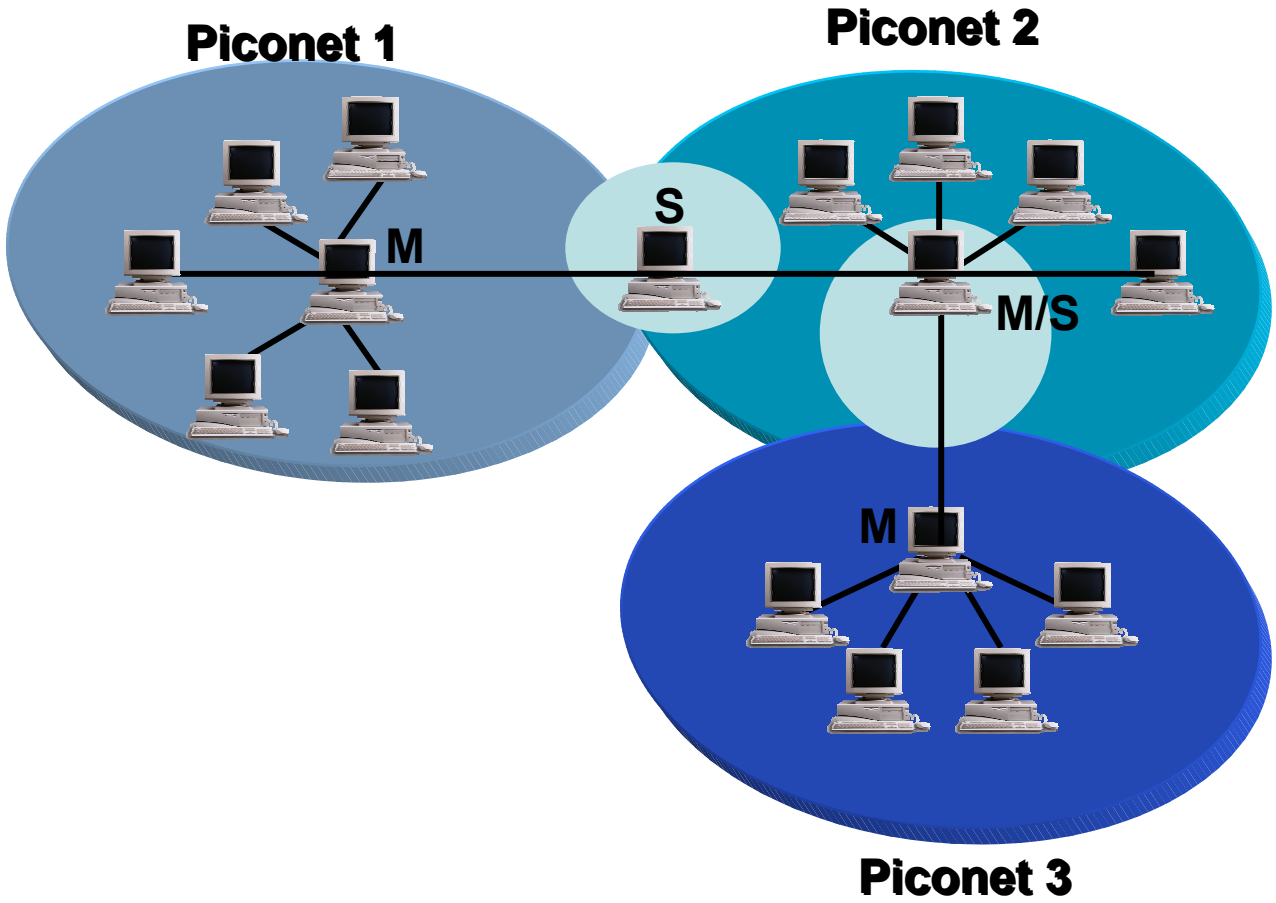
- Up to 8 active devices per bubble

79 channels of 1 Mbps

Simultaneous voice / data capable

- 432 Kbps (full duplex), 721/56 Kbps asymmetric
- or 3 simultaneous voice full duplex calls
- or a combination of voice & data

Bluetooth in a nutshell (4/4)



WiFi in a nutshell (1/4)

WiFi (Wireless Fidelity) is an extension of ethernet standard 802.3 managed by IEEE: 802.11

Roaming (peer to peer) or connect to a network (hot spots)

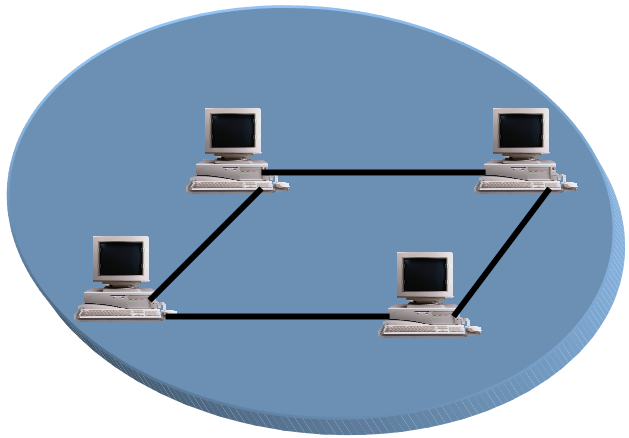
Several revisions but one worldwide reference: 802.11b

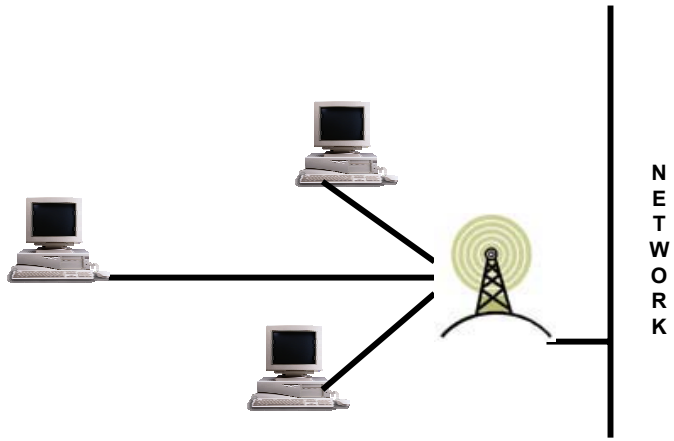
- Uses the open 2.4 GHz frequency range
- Range: 300m in open environments,100m in office environments
- Up to 11Mbps of data transfer

WiFi in a nutshell (2/4)

Two possible modes:

Ad Hoc





Infrastructure

WiFi in a nutshell (3/4)

Targeted devices: PC, Handhelds

Focused on Wireless performances:

- **Bandwidth**
- **Network architecture**
- **Huge industry sustaining**
- **Widely deployed**

WiFi in a nutshell (4/4)

All TCPIP applications are suitable:

- **Mails**
- **Voice over IP**
- **Games**
- **Remote control:**
  - Snapshot of PalmVNC 2.03



Pros & Cons (1/2)

- **Security: BT more secure than WiFi:**
  - BT Pairing (link keys), better authentication
- **Bandwidth:**
  - BT 720kbps / WiFi 11mbps
- **Capabilities:**
  - Bluetooth advertises capabilities, WiFi more complex
- **power requirement:**
  - BT: 1mW / WiFi: 1W

Pros & Cons (2/2)

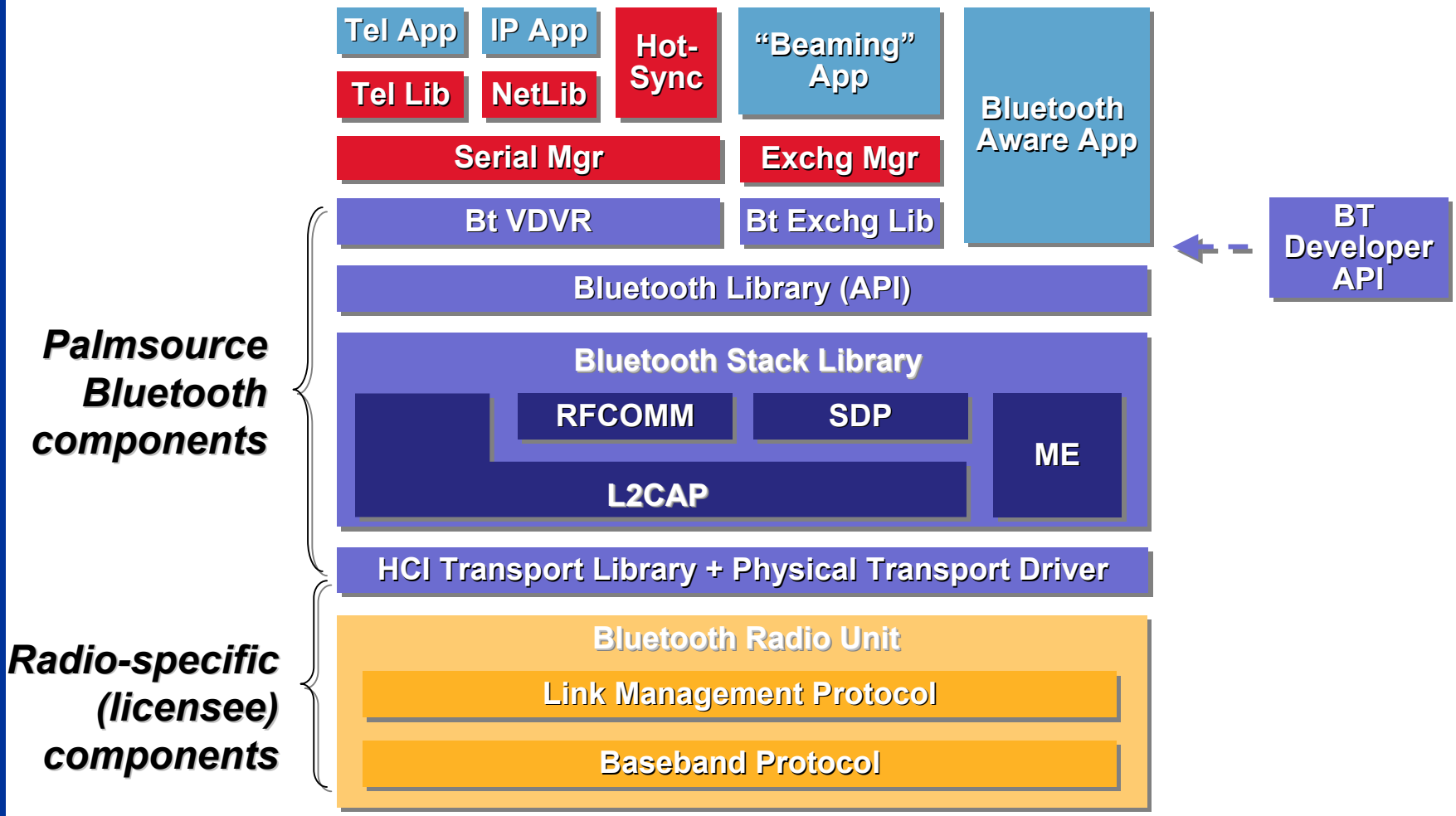
The Range:

- BTClass 1: 10m (1mW)
- BTClass 3: 100m (1W)
- WiFi: 100m (1W)

Enough space for both technologies:

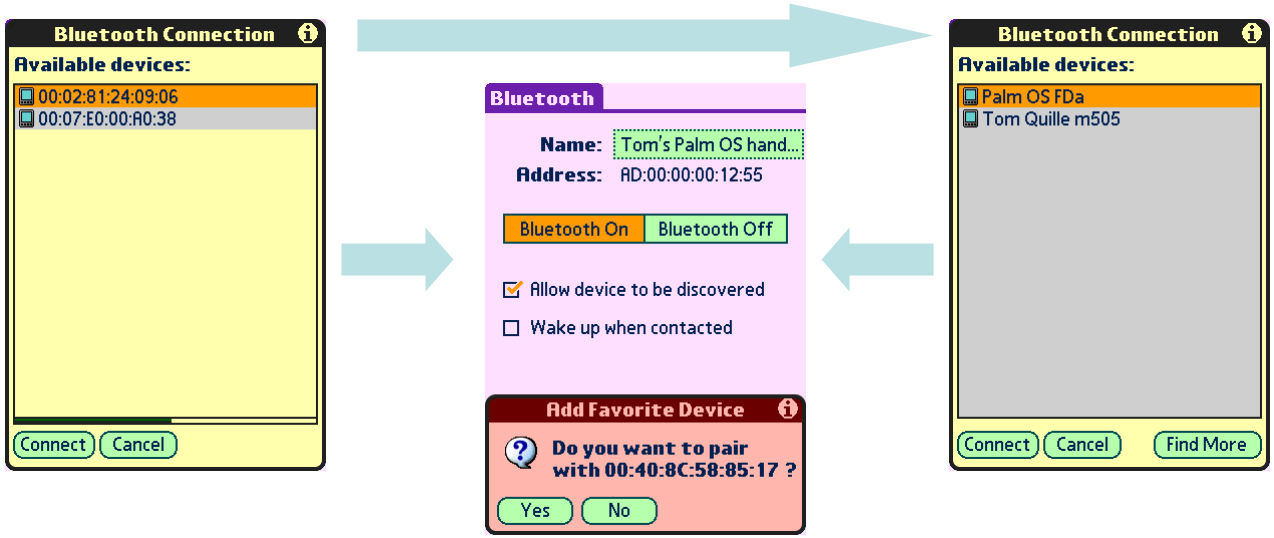
- **for example Microsoft:**
  - **Wifi for wireless LAN Access**
  - **BT for HID devices (keyboard and mouse)**

Bluetooth Architecture



Bluetooth: New Discovery UI

The current discovery UI is not very user-friendly, and is time-consuming… Palm OS 6 improves that, and an implementation of this UI, for previous systems is available from our web site…



Bluetooth: How to improve discovery UI

Using the standard BTLib API is only possible by licensees

BUT

You can use the basics API on ME level to address this improvement:

```
error = BtLibStartInquiry(gBtLibRefNum, 0, 255);
error = BtLibGetRemoteDeviceName(gBtLibRefNum,
   &(gDevices[giCurrentDevice].bdaddr), &gName,
   btLibRemoteOnly);
```

Bluetooth & Telephony (1/2)

Bluetooth in telephony: information exchange, dial, data transfers, headsets.

SCO channels (headset profile, handsfree profile) are not supported yet…

How to enable a voice connection: serial profile

- Use the virtual serial port to send an ATD command (or use TelSpcCallNumber)

Bluetooth & Telephony (2/2)

How to enable a data connection: Dial-up Networking profile

- Configure the connection panel
- This profile might be replaced by the PAN (Personal Area Network) profile but we still have one or two years before this

How to enable object exchange: Object push profile

- We only support the push feature and not the get or exchange one. But you can develop a new profile based on this feature: the BIP (Basic Imaging Profile) for example

Bluetooth: How to implement the BIP profile (1/2)

- **BIP profile:**
  - download or upload pictures
  - based on Obex
- **Two solutions:**
  - new exchange library
  - stand-alone application

- **In any case: introduce a new Obex stack, basic XML support and a JPG management**

Bluetooth: How to implement the BIP profile (2/2)

Example of code you could write with the PalmOS API:

- Advertise the profile:

```
err = BtLibSdpServiceRecordCreate( gBtLibRefNum,
   &sdpRecordH);
err = BtLibSdpServiceRecordSetAttributesForSocket (
   gBtLibRefNum, socket, (BtLibSdpUuidType*)
   &bipUUID, 1, appBtName, StrLen(appBtName),
   sdpRecordH);
err = BtLibSdpServiceRecordStartAdvertising( gBtLibRefNum,
   sdpRecordH)
```

- Register against the exchange manager

```
err = ExgRegisterDatatype (creatorID, exgRegExtensionID,
   «.jpg», «JPEG», 0)
```

Bluetooth: How to implement the PAN profile

- PAN might replace LAN and Dialup profiles in 2004
- Based on BNEP (Basic Network Encapsulation)
- Not yet in PalmOS, hard for developers to create:
    - Modify the Netlib
    - Write a new virtual serial driver

Wait for licensee solution

Bluetooth: How to implement the HID profile

- HID (Human Interface Device)is based on Basic GAP (General Access Profile)
- Runs natively on L2CAP
- Based on USB HID class devices (HID protocol)
- HID Devices: mice, joysticks, keyboards…

- Best way to go: a standalone application, using RFCom or L2CAP sockets, and emulating the protocol

WiFi: Palm Wifi SDK solution (1/2)

The Palm Wifi SDK is available on Palm site:

- You need to register to the PluggedIn Program

http://pluggedin.palm.com/regac/pluggedin/auth/SearchDocuments

(Download the Palm SG SDK v1.0)

- The SDK is enabling the Netservices API.

- Discover, connect and disconnect from 802.11 LAN spots

- Check and configure your WiFi solution

WiFi: Palm Wifi SDK solution (2/2)

The main Netservices APIs:

NetServicesStartScan	**NetServicesGetScanResults**
NetServicesSetRadioState	**NetServicesGetRadioState**
NetServicesGetPowerState	**NetServicesGetDeviceInfo**
NetServicesGetHWAddr	**NetServicesGetCommsQuality**
NetServicesProfileAdd	**NetServicesProfileDelete**
NetServicesProfileRead	**NetServicesProfileWrite**
NetServicesProfileFind	**NetServicesProfileSetOrder**
NetServicesProfileGetOrder	**NetServicesProfileGetInfo**
NetServicesGetChannel	**NetservicesGetChannelList**

WiFi: How to develop a WIFI application (1/2)

Source sample code to get some information:

```
UInt8  strength      = 0;
UInt8  quality       = 0;
UInt16 channel       = 0;
err = NetServicesGetChannel( gNetStat.libRef, &channel );
err = NetServicesGetCommsQuality( gNetStat.libRef,
              &strength, &quality);
```

Note: Signal strength and quality values are non-linear and may vary from device to device. Check the documentation

WiFi: How to develop a WIFI application (2/2)

Source sample code to do a BSS (Basic Service Set) scan:

Register for the notification sent when the scan is complete:

```
SysNotifyRegister(cardNo, dbID, WIFI_SCAN_COMPLETE_EVENT, NULL,
   sysNotifyNormalPriority, &gNetStat );
```

The notification has to be processed in PilotMain:

```
case sysAppLaunchCmdNotify:
   if (((SysNotifyParamType*)cmdPBP)->notifyType ==
        WIFI_SCAN_COMPLETE_EVENT ) { /* do something */ }
```

Start the scan:

```
err = NetServicesStartScan(gNetStat.libRef);
```

Retrieve the results asynchronously:

```
NetBssListType *bssListP = NULL;
err = NetServicesGetScanResults(gNetStat.libRef, NULL, &reqSize);
bssListP = MemPtrNew( reqSize );
err = NetServicesGetScanResults( gNetStat.libRef,
                (UInt8 *) bssListP, &reqSize );
```

WiFi: How to develop a WIFI HW solution

How to develop an Hardware solution for WiFi:

- **Either be a licensee and modify the Netlib consequently as well as the connection panel**

- **Or be a third party and go through the usb or serial port on PPP and integrate in HW module a PPP server and 802.11 stack (like the Xircom solution)**

Bluetooth : The Future

Bluetooth 1.2 (radio and baseband improvements, transparent to upper layers):

- **Faster connection**
- **Adaptive Frequency Hopping => reduce interference**
- **Enhanced Voice Processing and Quality of Service**
- **Anonymity Mode: mask physical address of a radio**

Bluetooth 2.0:

- **The software stack will evolve as well as the radio**
- **Better bandwidth…**
- **Specification in progress**

WiFi: The future

- 802.11g is starting to be deployed
- provides a 54Mbps bandwidth
- backward compatibility with 802.11b

Bluetooth & Wifi: best developers resources

BT SIG: www.bluetooth.com

The Wifi Alliance: www.wi-fi.org

IETF: www.ietg.org

Palm: pluggedin.palm.com

PalmSource:

- www.palmos.com/dev/tech/bluetooth
- Bluetooth and Communication forums

Peter Easton site: www.whizoo.com

Will be available to answer questions in Labs…

Thank You!